Chapter 20: Factor Analysis

1. Introduction

Exploratory Factor Analysis (EFA) identifies underlying latent constructs. Scree plots guide factor extraction; loadings indicate item-factor relationships.

2. Dataset

9 survey items from 300 respondents.

3. Methods

Principal Component Analysis performed; scree plot identified 3 factors. Factor Analysis conducted with maximum likelihood estimation.

4. Results

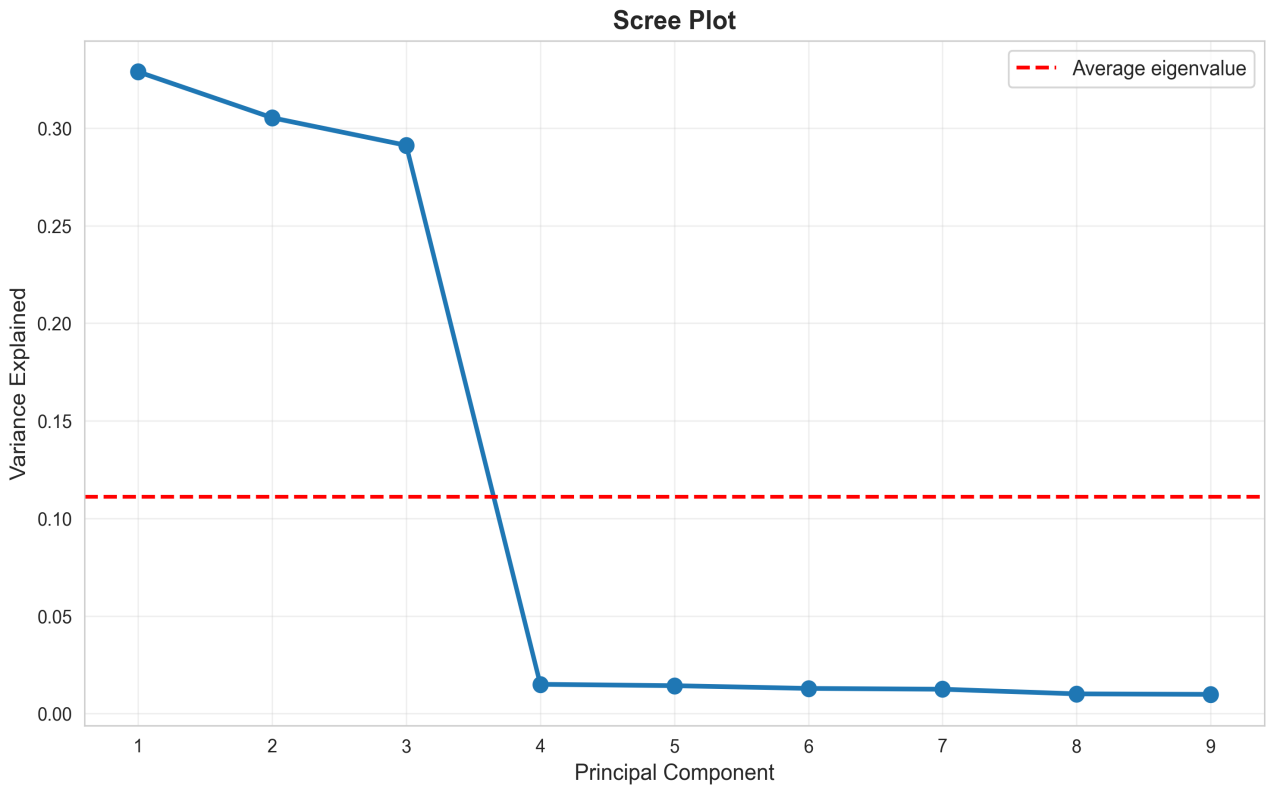

Fig 1: Scree plot shows variance explained by each component.

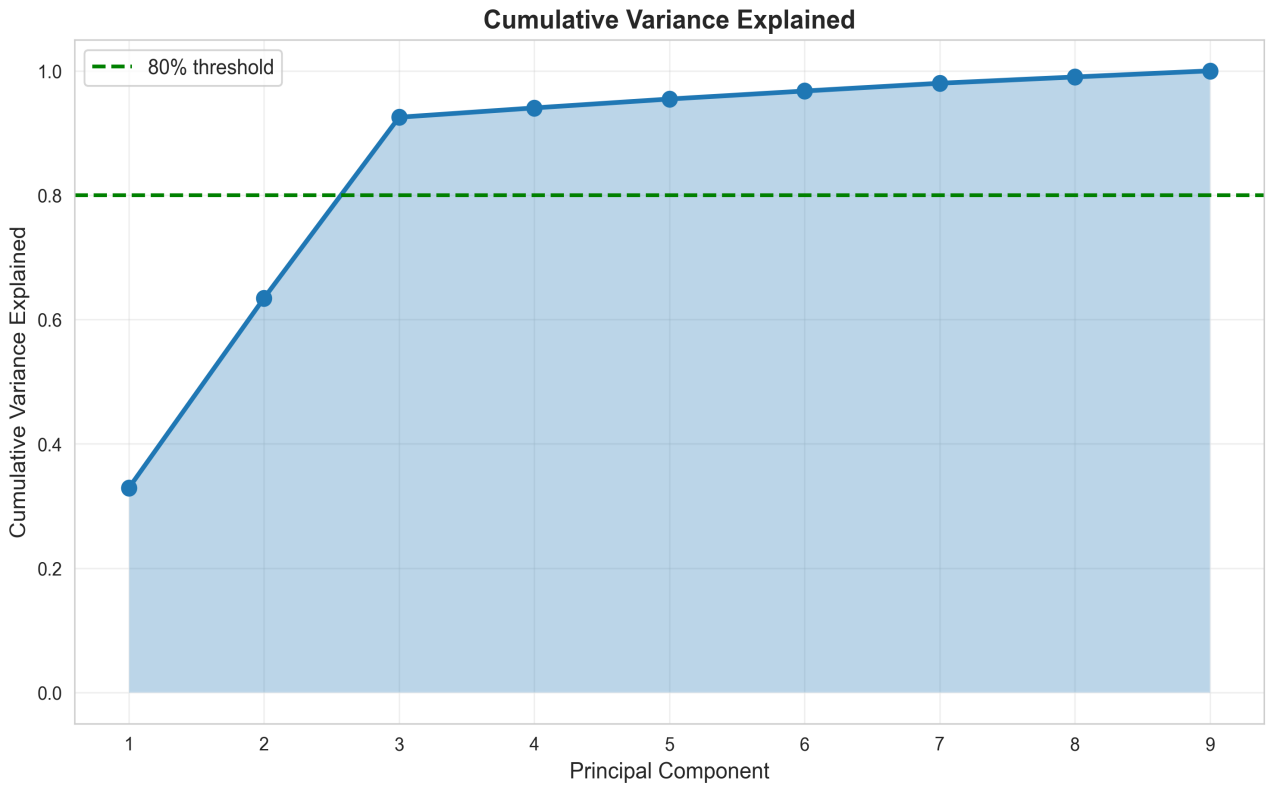

Fig 2: Cumulative variance explained shows information retained.

5. Interpretation

3 factors explain most variance. High loadings indicate strong item-factor associations. Factor interpretation comes from examining high-loading items.

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